

02 Prototype

DocShield AI — working web prototype, live and open source

1. Access — deployed application and source

Both links are live and require no credentials

Application <https://trust-doc-ai.vercel.app>

Repository <https://github.com/Hajji-ahmed/TrustDoc-AI>

Upload a PDF, JPG or PNG, run the analysis, and receive an authenticity pre-check: a risk score, anomalies drawn on the document itself, and a recommendation. The interface states plainly that the result is decision support — not an expertise, not a proof.

2. Pipeline

Upload → Normalization → Multimodal analysis → Deterministic checks → Risk score → Report

01	Upload	PDF, JPG or PNG up to 10 MB. Every page of a PDF is rendered, not only the first.
02	Normalization	Pages are rasterized client-side. Preview and model input are the same image, so a flagged region always lands where the reviewer looks.
03	Multimodal analysis	All pages go to the model in a single request, which lets it compare values across pages instead of judging each page alone.
04	Deterministic checks	Code re-verifies what a model cannot be trusted with: identifier length, IBAN mod-97 key, VAT arithmetic and rate, dates that do not exist, periods ending before they begin, values differing between pages.
05	Report	Risk score out of 100, numbered anomalies, extracted fields, and a recommendation oriented toward human verification.

3. Division of labour — what the model is never asked to do

Handled by the model

- ✓ **Reads** the document and extracts fields.
- ✓ **Sees** visual manipulation: mixed fonts, broken alignment, re-inserted text.
- ✓ **Compares** pages against one another.

Never trusted to the model

- ✗ **Counting digits** — identifier length.
- ✗ **Checksums** — IBAN mod-97 key.
- ✗ **Arithmetic** — VAT and column totals.

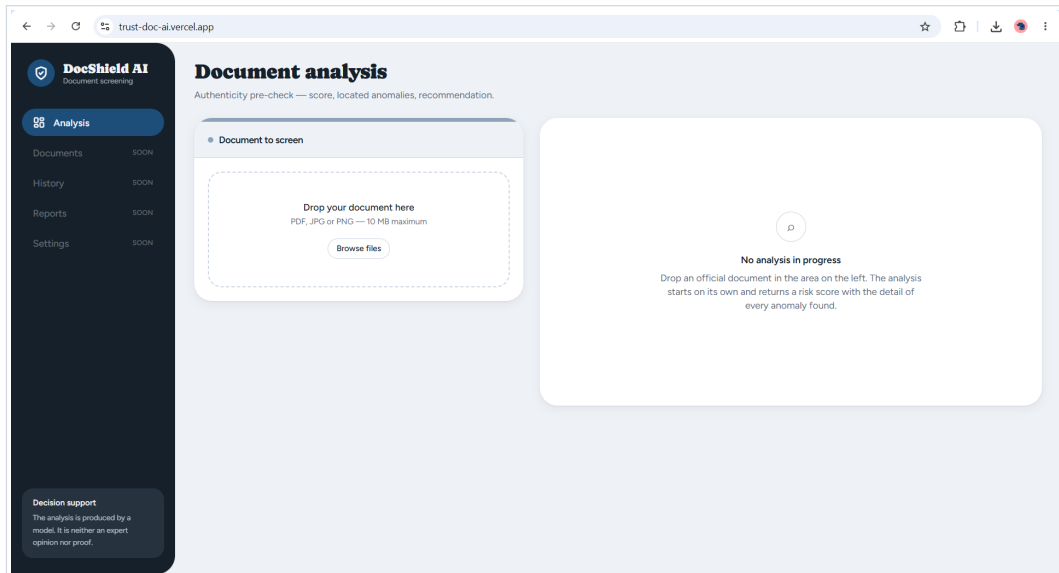
Deterministic code: exact, and auditable line by line.

A failed deterministic check raises the score floor and blocks an “accept” verdict. A model that saw nothing cannot overrule arithmetic that does not add up.

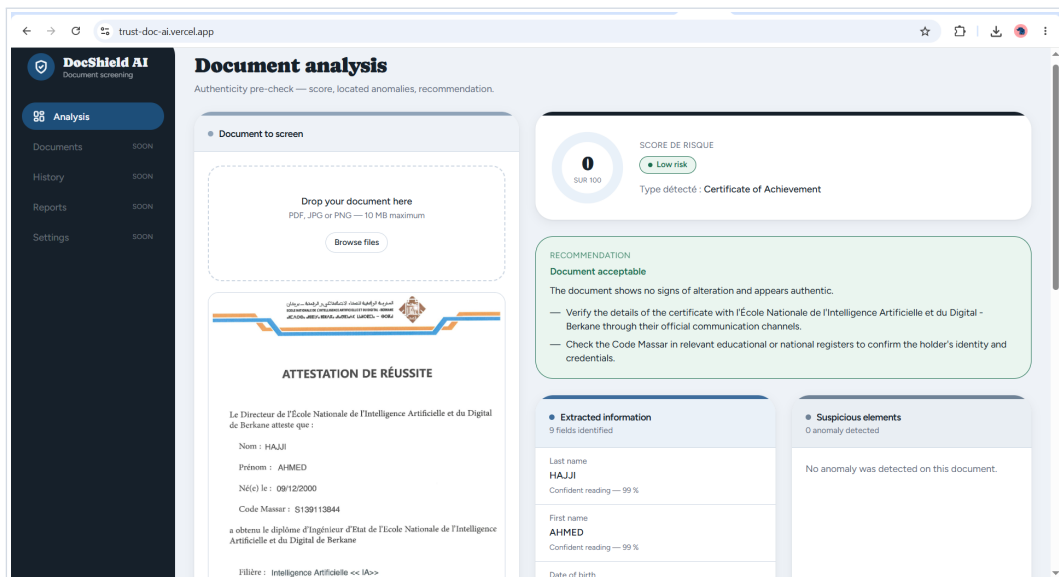
4. What the repository contains

app/api/analyze	Analysis endpoint. Validates the request, calls the model, re-validates the output, then consolidates it.
lib/prompt.ts	The system prompt: inspection method, score calibration, field-naming rules, and the ban on verifying a document against its own contact details.
lib/schema.ts	Strict JSON schema. A malformed analysis is rejected server-side, never half-displayed.
lib/validators.ts	The deterministic checks and the consolidation that keeps score, verdict and narrative from contradicting one another.
lib/pdf.ts	Multi-page rasterization, with the page cap that bounds cost per analysis.

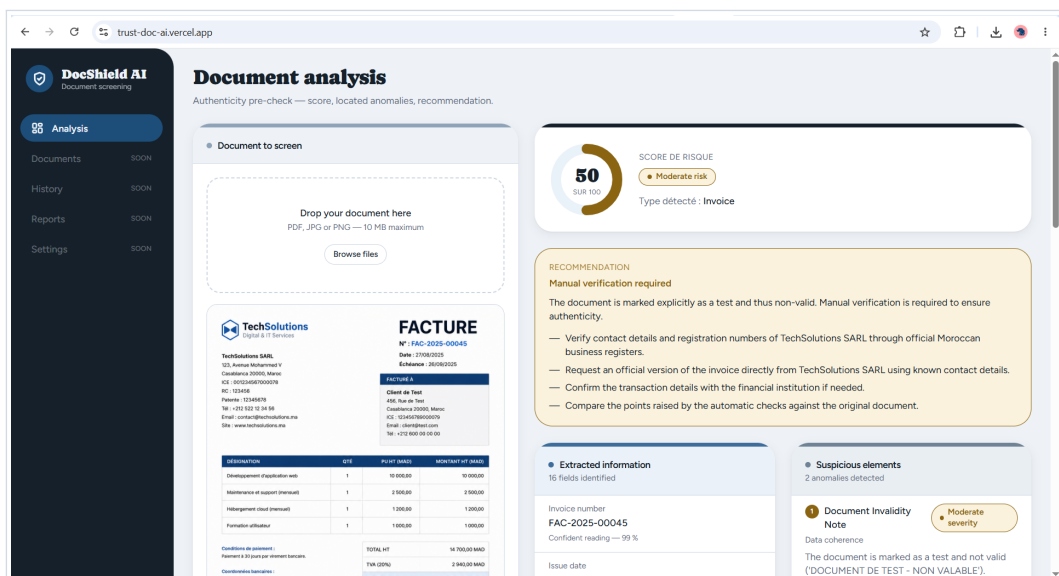
5. Screenshots



1. Initial state — the upload area, and a result panel that says what it is waiting for rather than showing an empty score.



2. A genuine certificate — no anomaly, and next steps pointing to the issuing school and to public registers rather than to the contact details printed on the document. The capture predates a later change: a clean document now scores a small residual instead of zero, because zero reads as a certificate of conformity and the tool certifies nothing.



3. An invoice at moderate risk — two anomalies, each localised on the document and numbered against the list, with the verification steps that follow from them.

6. What is proven, and what is not

The distinction matters more than the feature list. Everything on the left has been observed working; nothing on the right has been demonstrated, and none of it is claimed.

Proven

- ✓ Upload and rendering of PDF, JPG and PNG.
- ✓ Field extraction on every document tested, including a flattened scan.
- ✓ Anomalies localised and numbered on the page.
- ✓ Deterministic checks applied to exact rules, covered by automated assertions.
- ✓ Multi-page documents sent to the model in one request.
- ✓ Recommendations that route verification outside the document.

Not yet proven

- × Detection accuracy across fraud types.
- × Detection of a well-made or AI-generated forgery.
- × Cross-*document* verification within one file.
- × Metadata and file-history forensics.
- × Verification against an external issuer or register.
- × Behaviour at production scale.

These are the normal limits of a prototype built in 72 hours, and they are listed rather than left to be discovered. The first three are the subject of deliverable 03; the last three are architecture, not tuning.

7. Positioning

The sentence the whole product rests on

DocShield AI does not certify documents. It identifies suspicious signals and provides the evidence behind them, so that a human reviewer makes a better verification decision. A document that is fabricated but internally coherent passes every check here: screening detects inconsistency, not falsity.

Out of 72h scope, part of the target architecture: OCR tuning, dedicated manipulation models, metadata forensics, cross-document checks, issuer verification.